



OPEN Cloud-based slope risk monitoring and early warning system for open-pit coal mines: a case study of Zhonglian Runshi

Xiaoyong Dong¹, Shuang Li¹, Ran Ma^{1✉}, Weihui Tian¹, Keke Zhao¹, Hui Xiang¹, Jiongjiong Zhu¹ & Yang Qiu^{2✉}

Slope landslides in open-pit coal mines critically threaten mining safety. Traditional early-warning systems face limitations, including data silos, false alarms, and an overemphasis on data collection over analysis. To overcome these challenges, this study develops a novel cloud-based Early Warning System (EWS) that integrates the Internet of Things (IoT) for multi-source data acquisition, Long Short-Term Memory (LSTM) networks for high-precision time-series prediction, and Dempster–Shafer (D-S) evidence theory for uncertainty-based multi-indicator fusion. Unlike conventional EWSs relying on single-threshold rules, the proposed framework performs real-time, data-driven risk reasoning, significantly reducing false alarms and enhancing system robustness under heterogeneous monitoring conditions. Field deployment at the Zhonglian Runshi open-pit mine demonstrated that the proposed LSTM model achieved a prediction accuracy of $R^2 = 0.91$ ($RMSE < 0.11$ mm), while maintaining a tenfold lower computational cost than Transformer-based models. Through multi-source integration of 69 sensors, the system achieved continuous, real-time risk assessment with reliable multi-channel alerts. These results confirm the feasibility of lightweight, cloud-enabled intelligent EWSs for large-scale open-pit mines, providing a scalable paradigm for intelligent geohazard monitoring and early warning.

Keywords Open pit coal mine, Slope risk, Monitoring and early warning, Cloud-based system

Coal accounts for over 60% of energy consumption in China and serves as a cornerstone of the nation's energy security¹. Currently, more than 350 open-pit coal mines are in operation across China². Landslides on these unstable slopes pose a major safety threat, endangering the lives of miners, damaging production facilities, disrupting operational schedules, causing significant economic losses, and thus emerging as a critical hazard in mining safety. For example, the incident on February 22, 2023, at the open-pit mine of Xinjing Coal Industry in Alxa, Inner Mongolia, resulted in 53 fatalities, 6 injuries, and economic losses totaling 204 million yuan³. Developing highly reliable slope monitoring and early-warning technologies for open-pit mines is therefore essential to addressing these challenges and represents a key research focus in mining engineering.

Monitoring the mechanical response characteristics of slope rockmass under mining disturbances is a prerequisite for landslide early warning. Scholars have achieved the analysis and prediction of landslide risks by analyzing multi-source monitoring data. For instance, Li et al.⁴ deployed sensors including GNSS, piezometers, and rain gauges to monitor the mechanical response of an open-pit mine, providing substantial data support for landslide analysis. Cheng et al.⁵ utilized 'Space-Sky-Ground-Underground' integrated monitoring technology to enable real-time perception of multi-source mechanical responses in slope rockmass. However, on-site monitoring involves a multitude of devices, generating large volumes of data with varying collection intervals and structural formats. These issues pose significant challenges to data management and analysis, often leading to the formation of data silos⁶. Implementing advanced information technology to establish a monitoring and early warning system compatible with multi-source heterogeneous monitoring data is crucial for enhancing data management and mining efficiency. This represents another key technology for enabling effective early warning of slope hazards in open-pit mines.

The advancement of Internet of Things (IoT) technology has made real-time early warning for geological hazards like slope failures a tangible reality. Numerous scholars have successfully implemented various

¹Zhonglian Runshi Xinjiang Coal Industry Co., Ltd, Xinjiang, China. ²Center for Rock Instability and Seismicity Research, School of Resources and Civil Engineering, Northeastern University, Shenyang, China. ✉email: 1126491621@qq.com; 2210455@stu.neu.edu.cn

wireless communication technologies to form IoT networks, enabling the continuous collection and real-time transmission of critical monitoring data^{7–11}. As summarized in Table 1, each technology offers distinct trade-offs. For instance, LoRa is renowned for its long-range and Low-power consumption but suffers from a very low data rate, making it unsuitable for high-frequency data transmission. Zigbee supports flexible mesh networks but is limited by its short communication range. WiFi provides high bandwidth for rapid data transfer, yet its high-power consumption and limited coverage hinder its deployment in vast, remote areas.

In contrast, 4G technology stands out for its excellent balance of wide area coverage, high reliability, and substantial bandwidth, facilitating robust and near-real-time data transmission. This is particularly relevant in the context of China, where 4G infrastructure is extensively deployed and offers unparalleled network coverage, even in many remote and mountainous regions. Leveraging this existing, ubiquitous cellular network allows for a more straightforward, cost-effective, and reliable system implementation compared to deploying and maintaining a dedicated network infrastructure (like LoRa gateways or WiFi mesh). Therefore, for slope monitoring applications within areas with adequate 4G signal, directly utilizing 4G for data transmission presents a superior and pragmatic solution, ensuring seamless connectivity and timely hazard warnings.

Building upon IoT technology, scholars are now integrating cloud computing and other tools to develop disaster monitoring and early-warning systems. For example, LIU et al.¹² integrated various monitoring data to establish a comprehensive landslide monitoring system. Using this system, they achieved the identification of landslide risks and enhanced production safety. Bai D et al.¹³ designed a landslide monitoring and early warning system based on a microservices architecture, effectively resolving issues of module compatibility. Tao Z et al.¹⁴ developed an intelligent landslide monitoring mobile application (app), enabling timely and dynamic early warnings for geological hazards. Zhou et al.¹⁵ established a remote real-time monitoring system utilizing cloud computing technology and data transmission technologies, effectively processing data while addressing challenges such as the failure to achieve continuous monitoring.

Although many EWSs have been developed^{16,17}, most still rely on single-threshold or rule-based criteria, which are highly sensitive to sensor noise and often produce inconsistent or delayed warnings. These shortcomings mainly arise from the lack of effective multi-source data fusion, limited ability to model nonlinear deformation behaviors, and high computational cost for real-time applications.

To overcome these limitations, this study develops a cloud-based intelligent EWS that integrates IoT-enabled monitoring, LSTM-based deformation prediction, and Dempster–Shafer (D–S) theory for uncertainty-driven multi-indicator fusion. The proposed framework performs real-time data acquisition, temporal prediction, and probabilistic risk reasoning within a unified cloud–GIS platform, enabling dynamic, self-adaptive assessment of slope stability.

Field validation at the Zhonglian Runshi open-pit mine demonstrates that the system achieves a prediction accuracy of $R^2=0.91$ ($RMSE<0.11$ mm), while maintaining tenfold lower computational demand than Transformer-based networks. Overall, the proposed system establishes a lightweight and scalable paradigm for intelligent early warning in large-scale open-pit coal mines.

Designation of the early-warning system
Theoretical basis

The proposed EWS integrates four core technologies: Internet of Things (IoT), deep learning, cloud computing, and Geographic Information Systems (GIS). Each plays a distinct role in system development:

The IoT is an emerging technology for information sensing and transmission. It connects physical objects with digital networks using sensors and communication protocols, enabling intelligent monitoring and management. Recent IoT advances have significantly improved slope monitoring capabilities. These include multi-source data sensing and real-time transmission, which is crucial for accurate landslide monitoring and early-warning. In this research, IoT technology was implemented for the monitoring of open-pit mine slopes. Monitoring data were transmitted to cloud servers via efficient communication protocols (e.g., MQTT, HTTP), ensuring both data integrity and real-time availability. These real-time reliable multi-source monitoring data are essential for dynamic risk assessment and predictive early-warning of landslides.

Deep learning, a pivotal branch of artificial intelligence, demonstrates significant advantages in complex pattern recognition and prediction tasks. By constructing multi-layer neural networks, it automatically extracts intrinsic features and evolutionary patterns from massive datasets, enabling high-precision predictions for unknown data¹⁸. In geotechnical engineering, particularly for slope stability monitoring and early warning, this technology provides innovative solutions for capturing nonlinear, non-stationary data characteristics that traditional methods often fail to address¹⁹. This study employs Long Short-Term Memory (LSTM) algorithms to analyze temporal evolution patterns from slope monitoring data, establishing an advanced prediction model for slope deformation.

Scholar	Monitoring Target	Technology	Advantages	Limitations
Dorthi et al. ⁷⁸	Slopes over old underground galleries	Zigbee	Low power, Mesh networking	Short range,
Yadav et al. ⁹	Open-cast mine slope	LoRa	Long range, Low power	Low data rate, Limited network capacity
Marino et al. ¹⁰	Landslide-prone areas	WiFi	High bandwidth, Fast transmission	High power consumption, Limited coverage
Ciampalini et al. ¹¹	Ground deformations of the slope	4G	Wide coverage, High reliability, High bandwidth	Operational cost

Table 1. Comparison of representative IoT-based slope monitoring systems.

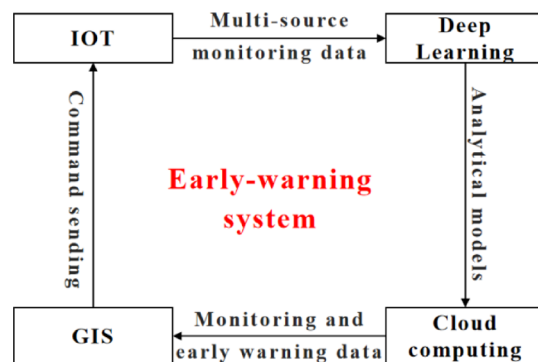


Fig. 1. Technology composition of the early-warning system.

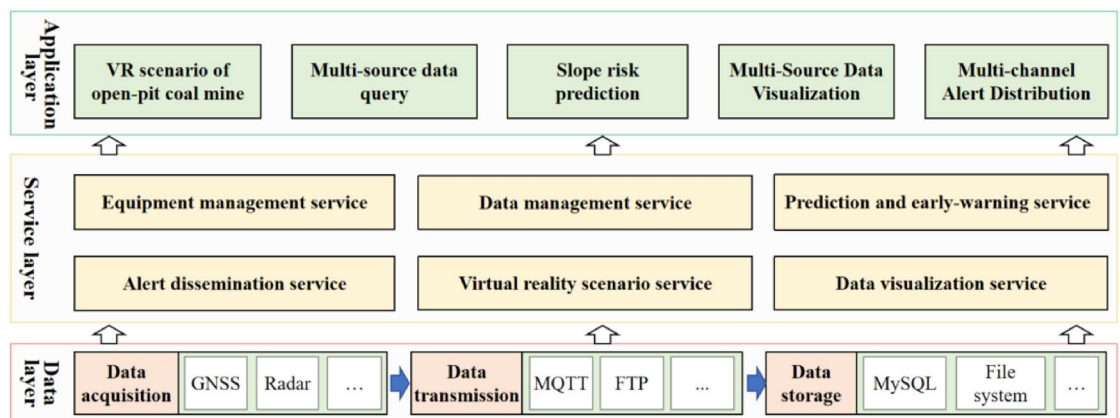


Fig. 2. Framework of the EWS.

Cloud computing represents an advanced computational mode that creates configurable resource pools. These pools provide on-demand access to computing resources (networks, servers, storage, applications, and services) anytime, anywhere. This mode significantly reduces resource management workloads and minimizes interaction costs with service providers. It enables rapid resource deployment and release²⁰. This study leverages Alibaba Cloud's Elastic Compute Service (ECS), a virtual server solution in cloud environments. ECS dynamically adjusts computing resources based on user requirements, ensuring efficient resource utilization. By integrating ECS's advantages (elastic scaling, efficient management, and high reliability), this research achieves two critical functions: (1) Storage of multi-source monitoring data. (2) Real-time data analysis. These capabilities deliver robust computational support for landslide risk assessment and early-warning.

GIS is an integrated system designed for data acquisition, transmission, management, storage, analysis, and visualization. Their core strength lies in efficient integration and dynamic representation of geospatial data²¹. Recent advances in 3D modeling, virtual reality (VR), and artificial intelligence have significantly expanded GIS applications in geotechnical engineering. These technologies demonstrate particular advantages in slope stability monitoring and landslide early-warning. In this study, GIS capabilities for multi-source data management and visualization were leveraged to develop a virtual reality scenario for slope risk management. By integrating topographic, monitoring, and early warning data, the system provides a foundation for mine safety management.

The interrelationships among the four technologies are illustrated in Fig. 1. The EWS acquires multi-source monitoring data through IoT technology. Deep learning provides analytical models for data mining, while cloud computing delivers computational power for data analysis and landslide risk assessment. GIS technology enables integrated visualization of real-time monitoring and early warning data, achieving transparent risk representation. Finally, users manage risks via the GIS system and dispatch control commands through IoT technology, establishing closed-loop management of landslide risks.

Framework design for the EWS

The EWS employs a three-tier framework (Fig. 2): data layer, service layer, and application layer. The data layer is responsible for the real-time collection and storage of on-site monitoring data. The service layer handles data processing to meet the requirements of various disaster early-warning procedures. The application layer is dedicated to the visualization of the early-warning process. This structure ensures comprehensive management of the entire landslide risk process – from data acquisition and risk analysis to warning issuance and risk visualization. Key design considerations for each tier are detailed below.

Data layer

The data layer manages real-time acquisition, transmission, and storage of multi-source slope mechanical response data. This EWS integrates multiple monitoring devices for open-pit slopes. These include GNSS devices, multi-point extensometers, inclinometers, and radar for tracking rockmass displacement. Additionally, crack gauges measure crack width. Blast vibration sensors detect shockwaves from mining operations. These real-time, multi-source monitoring data provide the foundation for comprehensive landslide risk assessment and early warning. Given the significant variations in data structures and acquisition approaches across monitoring devices, the system incorporates tailored data transmission schemes designed for the specific characteristics of each device type to enhance efficiency:

- (1) Point-based monitoring devices, such as GNSS devices and inclinometers, generate multi-channel time-series data. This data type is characterized by small individual packet sizes but high acquisition frequencies. To efficiently transmit this data, the system employs the MQTT protocol. MQTT demonstrates high performance for small payloads, high-frequency communication, and reliable data delivery, making it well-suited for the real-time transmission needs of point-based sensors.
- (2) Radar monitoring offers extensive coverage, high scan frequency, and generates large data volumes per measurement. Following each scan, the radar device locally produces a TXT file containing deformation data. To enable efficient transmission of these substantial files, the EWS employs a combined approach using file system monitoring and the FTP protocol. The file system monitor detects newly generated TXT files in real-time, triggering their immediate transfer to a designated server via FTP. This method ensures both prompt data delivery and file integrity.

Following data acquisition and transmission, the EWS implements differentiated storage strategies based on data types to ensure efficient management and rapid retrieval. For structured data from GNSS and crack gauge sensors, MySQL databases provide optimized storage, leveraging their robust data management and query capabilities. Regarding file-based data generated by radar systems, characterized by large file sizes, the system stores files directly on local disks while using MySQL to record storage paths and metadata. This hybrid approach reduces database load and improves file management efficiency.

Furthermore, to ensure reliable multi-source data transmission and real-time communication, the IoT network within the proposed EWS adopts a 4G-based wireless architecture. Each on-site monitoring device—including GNSS, radar, inclinometers, and crack gauges—is equipped with a data acquisition module and a 4G communication terminal. Monitoring data is transmitted through the public 4G cellular network to the cloud server, where the data layer receives and synchronizes information in real time.

Building on the IoT architecture studies by Dorthi²² and Zhou¹⁵, the IoT communication framework adopted in this study follows a three-tier structure comprising the perception layer, network transmission layer, and application layer.

The perception layer comprises distributed monitoring sensors that collect displacement, deformation, and vibration data from slopes in real time. The network transmission layer utilizes 4G communication protocols to achieve long-distance, low-latency, and stable data transmission between field devices and the cloud platform. The use of existing 4G infrastructure ensures high reliability and scalability while minimizing deployment cost. The application layer runs on the cloud platform, integrating real-time data acquisition, storage, and analysis services through the MQTT and HTTP protocols.

To visually clarify the IoT communication process, a network architecture diagram has been added to the revised manuscript (Fig. 3). Monitoring sensors (GNSS, radar, inclinometer, and crack gauge) collect real-time deformation data and transmit them wirelessly to the 4G base station. Using MQTT and HTTP protocols, the base station uploads the data through the 4G network to the cloud platform for data storage, analysis, and visualization. The process ensures real-time synchronization, low latency, and high reliability for continuous slope risk monitoring.

Service layer

The service layer functions as the core component of the EWS, enabling intelligent data processing and analysis to support all phases of landslide hazard alerts. It comprises six integrated modules: (1) Equipment management services; (2) Data management services; (3) Prediction and early-warning service; (4) Alert dissemination service; (5) Virtual reality scenario service; (6) Data visualization service. These modules operate collaboratively

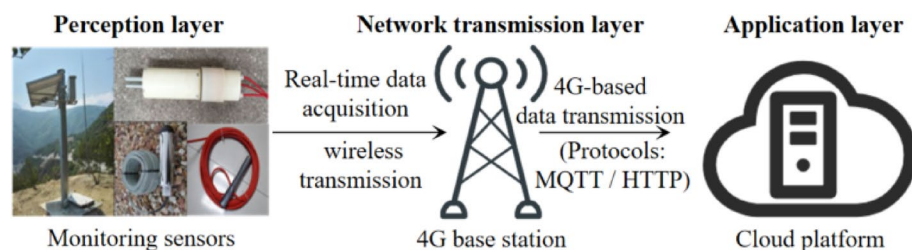


Fig. 3. 4G-based IoT architecture for EWS.

to establish a comprehensive monitoring and warning framework for open-pit coal mine slopes. The specific functions of each module are detailed below.

- (1) *Equipment management service*: This service provides full lifecycle management for monitoring devices. Key functions include maintaining essential device information. By intelligent device-system matching, the service ensures real-time data transmission and cloud synchronization. This guarantees continuous and complete data acquisition, delivering reliable data for data management services.
- (2) *Data management service*: This service handles the storage, maintenance, and retrieval of multi-source monitoring data. It provides robust storage and efficient management capabilities for multi-source data streams. By establishing an integrated data management platform, the service enables standardized processing and rapid querying of monitoring datasets. This infrastructure delivers a solid data foundation for predictive analytics and early-warning services.
- (3) *Prediction and early-warning service*: This service delivers intelligent forecasting of rockmass mechanical behavior and slope stability. It integrates three core functions: data preprocessing, LSTM-based mechanical response prediction, and Dempster-Shafer evidence theory-driven landslide risk assessment. This integrated framework extracts critical landslide precursors from multi-source monitoring data. By implementing data fusion methods, it resolves conflicts in multi-device warning outputs, significantly improving the accuracy and reliability of slope risk predictions. The service thus provides a quantifiable basis for early-warning decision-making.
- (4) *Alert dissemination service*: This service manages early-warning alert distribution and administration. It enables flexible configuration of alert dissemination channels and closed-loop management of response procedures. By continuously monitoring slope risk status, the service automatically issues alerts to management personnel when the slope is instability. All response actions are systematically recorded, ensuring automatic alert distribution and full traceability of risk mitigation processes. This provides critical decision support for landslide management.
- (5) *Virtual reality scenario service*: This service constructs a digital twin scenario of the open-pit coal mine using the SuperMap platform. This system integrates UAV-based oblique photogrammetry to achieve high-precision modeling of coal mine slopes. By visualizing rockmass mechanical responses, the service transparently displays real-time slope conditions and deformation trends. This provides managers with an immersive virtual reality (VR)-enabled visualization experience for operational decision-making.
- (6) *Data visualization service*: This service converts multi-source monitoring data into computer-readable charts and graphs, significantly simplifying data interpretation. It dynamically displays real-time data while generating historical trend plots. These visual tools help users intuitively understand slope risk conditions. Through interactive visual exploration, the service enhances both the efficiency of slope risk management and the scientific basis for decisions.

Application layer

The application layer serves as the user interface for the EWS. It performs two key tasks: calling services and displaying functions. Designed around user needs, it combines multi-source data, intelligent analysis, and visual tools. This layer manages the full lifecycle of landslide early-warnings for open-pit coal mines, which delivers five primary capabilities:

- (1) *VR scenario of open-pit coal mine*: Leveraging device management and virtual reality scenario services, this module constructs high-precision reality models using UAV oblique photogrammetry. It enables precise device localization and dynamic visualization within a VR scenario.
- (2) *Multi-source data query*: A data center is built using the device management and data management services. Users can quickly retrieve monitoring data for specific timeframes and locations. This enables a better understanding of rockmass failure progression in slopes.
- (3) *Slope risk prediction*: This module integrates data management and Prediction and early-warning services to deliver intelligent risk assessment and prediction. The EWS processes real-time monitoring data through an LSTM-based prediction model. The output is then fused with D-S theory to evaluate landslide risk. This integrated approach proactively identifies potential landslide risks, providing a foundation for mitigation strategy development.
- (4) *Multi-Source Data Visualization*: This module integrates data visualization and virtual reality scenario services to provide a multi-view representation of multi-source monitoring data. By integrating diverse visualization formats (data tables, trend curves, and heatmaps), the EWS dynamically displays current slope conditions and evolutionary trends. This module streamlines data interpretation while delivering intuitive decision-making support to management personnel, which can significantly enhance emergency response efficiency.
- (5) *Multi-channel Alert Distribution*: This module integrates predictive analytics and alert dissemination services to generate smart tiered alerts for slope risks. Based on predefined risk classification protocols, it automatically disseminates alerts through SMS, email, and other communication channels. This ensures the timely delivery of critical notifications to relevant personnel, enabling rapid emergency response.

Early-warning methods

Prediction and early-warning methods constitute the core of the Early Warning System (EWS) and are crucial for generating high-precision landslide alerts. As shown in Fig. 4, the integrated workflow encompasses data processing, prediction, and risk assessment. The process begins with the real-time acquisition of rock mass deformation data via monitoring equipment such as radar and GNSS. A scheduled task mechanism, whose

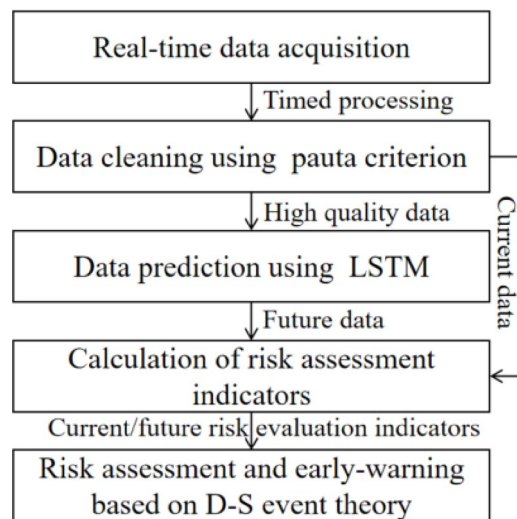


Fig. 4. Overall workflow of the early-warning method.

frequency is adaptively adjusted based on the identified landslide risk level, is employed for periodic data processing. By default, data are processed once per day, while the frequency increases to once per hour when the risk level reaches “Orange” or higher. Upon triggering, the scheduled task first performs data cleaning using the Pauta criterion, as detailed in Section “[Data preprocessing based on the Pauta criterion](#)”, to ensure high-quality monitoring data. The cleaned data subsequently serve two purposes: they are fed into an LSTM model to predict future rock mass deformation (Section “[LSTM-based prediction model](#)”), while both the cleaned data and prediction results are jointly analyzed to extract landslide risk evaluation indicators, such as the improved tangent angle, for both current and future states (Section “[Landslide risk assessment model based on D-S theory](#)”). These indicators are then fused via the Dempster-Shafer (D-S) evidence theory model, enabling a comprehensive landslide risk assessment and graded early warning, as further elaborated in Section “[Landslide risk assessment model based on D-S theory](#)”. Detailed methodological descriptions are provided in the subsequent sections.

Data preprocessing based on the Pauta criterion

Harsh mining environments with heavy dust and poor network connectivity often prevent monitoring devices from operating consistently. This instability compromises data integrity and reliability.

Incomplete monitoring data primarily stems from transmission anomalies during collection, caused by unavoidable factors like network disruptions, adverse weather, or mining activities. Resulting data gaps alter dataset structures—for example, shortening time-series sequences. Since most analytical algorithms require structurally consistent inputs, such disruptions frequently cause analytical failures, undermining slope risk assessment accuracy.

Given that most devices follow a scheduled acquisition time interval, data completeness is verified against predefined intervals. Let d represent the data measurement interval (in minutes). Every minute, the system initiates a missing-data check. For each check, the time difference (Δd) between the current check time and the device’s last successful data acquisition is calculated. When $\Delta d > d$, the device has failed to collect data within its scheduled measurement interval. To address such missing values, the LSTM-based prediction model (Section “[LSTM-based prediction model](#)”) is used to predict current mechanical responses. These predicted values are then stored as monitoring data to preserve dataset integrity.

Data unreliability mainly involves outliers from complex environmental interference during collection. These anomalies distort distributions and impair deep learning model performance. Consequently, robust preprocessing is essential for noise removal. The Pauta criterion (3σ principle)²³ provides effective denoising using statistical properties of normal distributions, combining operational simplicity with proven efficacy. The Pauta criterion is employed to identify noise by establishing a confidence interval based on the mean (μ) and standard deviation (σ) of a dataset, as defined in Eq. (1):

$$[\mu - k\sigma, \mu + k\sigma] \quad (1)$$

where k is a scaling factor that determines the breadth of the interval. When $k = 1$, approximately 68.27% of the data points are expected to lie within the interval; this proportion increases to about 95.45% for $k = 2$, and 99.73% for $k = 3$. Any data point falling outside this defined interval is subsequently identified and treated as noise. This study implements real-time Pauta denoising within a 7-day moving window—optimized for slope mechanical response characteristics. As new data arrives, the system continuously performs anomaly detection as follows:

1. For a real-time data point A_t measured at time t , extract data from the preceding 7 days to form a time-series dataset $L = \{A_0, A_1, A_2, \dots, A_t\}$.

2. Calculate the standard deviation (σ) and mean (μ) of dataset L . Determine the confidence interval C for normal data using Eq. (1). Following the Pauta criterion and considering data characteristics, a conservative threshold $k=2$ is adopted, retaining 95.45% of normal data for subsequent calculations.
3. Evaluate A_i against the confidence interval. Data points within C are classified as normal, otherwise as outliers.

LSTM-based prediction model

Current early-warning systems (EWSs) typically assess slope stability by comparing real-time monitoring data against predefined thresholds. While this approach is effective for evaluating instantaneous stability conditions, it lacks predictive capability and often leads to delayed warnings. In open-pit mines, slope monitoring data generally exhibit nonlinear and nonstationary temporal patterns. Despite their complexity, these deformation sequences frequently contain short-term trends that can be exploited through time-series modeling.

Long Short-Term Memory (LSTM) networks are well-suited for this task because their gated memory structure effectively captures nonlinear temporal dependencies in sequential data²⁴. After data denoising using the aforementioned procedures, the LSTM model is adopted in this study to forecast short-term mechanical responses, thereby providing near-future deformation trends. The workflow consists of four main stages:

- (1) *Pre-processing phase*: In this study, multiple types of monitoring instruments were deployed, including Global Navigation Satellite System (GNSS) sensors, ground-based radar, inclinometers, and crack gauges. The sampling intervals were 10 min for GNSS, 30 min for radar, and 5 min for inclinometers and crack gauges. The monitoring data are collected continuously and transmitted to the cloud server in real time. Given that monitoring devices collect data at intervals ranging from minutes to hours, their measurements often exhibit significant fluctuations. To mitigate these fluctuations, daily mean deformation values are calculated at the end of each day. These mean values represent the mechanical response of the slope rockmass for those 24 h.
- (2) *Model architecture*: The proposed LSTM model employs a concise univariate architecture designed for efficient and stable slope displacement forecasting. Each input sample consists of the most recent 10 consecutive daily displacement values, corresponding to an input shape of (10, 1)—ten time steps and one feature per step. The network includes three layers:
 1. an input layer receiving the displacement sequence.
 2. a single LSTM hidden layer with 100 memory units, which captures the nonlinear temporal evolution of slope deformation.
 3. a fully connected output layer with one neuron and a linear activation, producing the predicted displacement for the next day (output dimension = 1).

The LSTM layer uses the standard tanh and sigmoid activation functions for internal gating. This lightweight configuration provides sufficient representational capacity for long-term deformation sequences while maintaining low computational cost, enabling real-time operation on industrial hardware.

- (3) *Training phase*: For model training, the most recent 40 days of monitoring data are used. A sliding window of 10 days serves as the input, and the subsequent day's displacement as the target output. By moving the window forward one day at a time, 30 training samples are generated. The model is trained using the Adam optimizer (learning rate = 0.001) with mean squared error (MSE) as the loss function. Training proceeds for up to 100 epochs with a batch size of 10, reserving 10% of the samples for validation. An early stopping strategy (patience = 10, restore_best_weights = True) is applied to prevent overfitting. After each forecast, the predicted value is appended to the time sequence, forming a recursive prediction loop for continuous updating.
- (4) *Prediction phase*: During inference, the model receives the latest 10 days of actual monitoring data to forecast the next day's displacement. For subsequent predictions, the model combines the most recent 9 days of real data with the previous prediction to form a new input window, iteratively generating forecasts one day at a time. This process continues until 10 consecutive daily forecasts are obtained, providing a near-term prediction horizon for slope deformation evolution.

Landslide risk assessment model based on D-S theory

Current risk assessment practices for slopes in open-pit coal mines often rely on evaluating single parameters from monitoring data. However, slope failure involves complex mechanisms where analysis based on a single parameter is inadequate, frequently leading to conflicting risk assessments. To address this limitation, we apply Dempster-Shafer (D-S) evidence theory. This systematic framework integrates uncertain information from multiple sources. As an extension of probability theory, D-S quantitatively combines evidence from different indicators while accounting for their individual reliability, thereby resolving contradictions between multi-source assessments. This integrated approach provides a more comprehensive and accurate stability evaluation for slope risk management.

- (1) Evaluation indicator selection and quantification of risk value.

When a slope is undergoing creep deformation, Xu et al.²⁵ proposed that the displacement rate (v) during the secondary creep stage remains constant. To transform the displacement–time curve's vertical axis into the same time dimension as the horizontal axis, displacement is divided by v , as shown in Eq. (2):

Warning level	Warning form	Improved tangential angle (°)	Cumulative displacement (mm)	Deformation velocity (mm/d)
Advisory	Blue	$\alpha \approx 45$	$20 \leq S < 30$	$4 \leq v < 8.5$
Caution	Yellow	$45 \leq \alpha < 80$	$30 \leq S < 40$	$8.5 \leq v < 13$
Warning	Orange	$80 \leq \alpha < 85$	$40 \leq S < 50$	$13 \leq v < 17.5$
Alert	Red	$\alpha \geq 85$	$S \geq 50$	$v \geq 17.5$

Table 2. Comprehensive slope stability evaluation indicator system.

Evidence source	Unstable State	Stable State
Improved tangent angle (m_1)	R_α	$1 - R_\alpha$
Cumulative displacement (m_2)	R_S	$1 - R_S$
Deformation velocity (m_3)	R_v	$1 - R_v$

Table 3. Mass function allocation of evidence source.

$$T = \frac{\Delta S}{v} \quad (2)$$

where ΔS is the slope displacement. v is the secondary creep velocity, and T is the transformed vertical coordinate value with the same dimension as time. The tangent angle α of the displacement curve is then defined as the improved tangent angle, expressed as:

$$\alpha = \arctan \frac{\Delta T}{\Delta t} \quad (3)$$

When $\alpha < 45^\circ$, the rockmass is in the primary creep stage, and failure is unlikely. When $\alpha = 45^\circ$, it enters the secondary creep stage. When $\alpha > 45^\circ$, it enters the tertiary creep stage, indicating a potential risk for failure. This indicator (α) has been widely adopted by researchers internationally^{26,27}. Its effectiveness has been validated through extensive engineering practice, leading to the development of a corresponding four-level early-warning threshold.

Furthermore, based on the report analysis and literature review, cumulative displacement and deformation velocity were selected as additional evaluation indicators. Warning level thresholds for these indicators were established, thus forming a comprehensive slope stability evaluation indicator system. The specific thresholds are detailed in Table 2.

To further quantify the relationship between the different warning levels and the landslide risk, probability ranges for each level were assigned based on previous research²⁸, specifically 0 ~ 0.25 for Advisory, 0.25 ~ 0.5 for Caution, 0.5 ~ 0.75 for Warning, and 0.75 ~ 1 for Alert. Furthermore, relationships linking the landslide risk to each indicator were established through a fitting approach, with the specific mathematical expressions provided in Eqs. (4) to (6). This quantification thus enables enhanced landslide risk assessment and prediction.

$$R_\alpha = 0.079e^{0.0249\alpha} \quad (4)$$

$$R_S = 0.025S - 0.5 \quad (5)$$

$$R_v = 0.0556v - 0.2222 \quad (6)$$

(2) Landslide risk real-time assessment based on D-S theory.

Building upon the above framework, once new real-time monitoring data is received or a prediction is completed, the improved tangent angle, cumulative displacement, and deformation velocity are calculated. The risk level for each indicator is then quantified using Eqs. (4)–(6). Subsequently, the Dempster-Shafer (D-S) evidence theory is applied to fuse the risk parameters derived from these indicators. The procedure is as follows:

In this study, the risk state of the slope is defined within the identification framework $\Theta = \{A, B\}$, where A represents an unstable state and B represents a stable state. The risk assessment results for the improved tangent angle (R_α), cumulative displacement (R_S), and deformation velocity (R_v) serve as the evidence sources. Based on Eqs. (4)–(6), the basic belief assignments are determined. Mass functions describing the slope stability states are then constructed for each evidence source according to the principles outlined in Table 3.

Based on D-S evidence theory, the three independent mass functions (m_1, m_2, m_3) are fused into a combined mass function using Dempster's combination rule to obtain the comprehensive probability, R_f , of the slope being in an unstable state (Proposition A). The rule is defined as follows:

$$R_f = (m_1 \oplus m_2 \oplus m_3)(A) = \frac{1}{K} \sum_{A_1 \cap A_2 \cap A_3 = A} m_1(A_1) \cdot m_2(A_2) \cdot m_3(A_3) \quad (7)$$

Warning level	Warning form	R_f
Advisory	Blue	0 ~ 0.25
Caution	Yellow	0.25 ~ 0.5
Warning	Orange	0.5 ~ 0.75
Alert	Red	0.75 ~ 1

Table 4. Classification of slope risk warning levels.

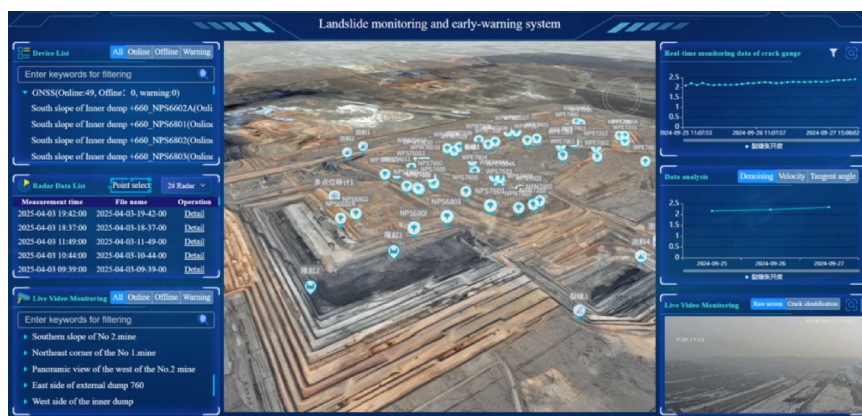


Fig. 5. Interface of the developed EWS.

The normalization factor K ensures that the total probability is unity and is defined as the sum of the products of masses for all non-conflicting propositions:

$$K = \sum_{A_1 \cap A_2 \cap A_3 \neq \emptyset} m_1(A_1) \cdot m_2(A_2) \cdot m_3(A_3) \quad (8)$$

The value R_f is subsequently classified into a risk level based on Table 4, achieving a data-driven and fused risk assessment.

Implementation of EWS

The EWS employs a Browser/Server (B/S) architecture. The frontend, developed with Vue.js, Element UI, ECharts, and Cesium, enables real-time monitoring data visualization and slope stability assessment visualization. The backend integrates Maven, Spring Boot, MyBatis, and Python for data management and intelligent analysis. Cloud infrastructure utilizes Aliyun ECS elastic servers and Aliyun RDS MySQL. This architecture eliminates client-side installations—users access the system instantly via any network-connected browser. Domain-based access initiates HTTP requests for frontend-backend data exchange, enabling one-click retrieval and visualization of slope risk state. Figure 5 presents the interface of the developed system.

Application analysis of the EWS

Mine overview

The Zhonglian Runshi coal mine is located approximately 90 km northeast of Qitai County, Xinjiang Province, China, covering an area of 12.884 km², as shown in Fig. 6. The mine area has high temperatures and low rainfall, with an absolute altitude of +670 to +710m.

At present, the open-pit mine has developed a pit with an excavation depth of approximately 90m, along with one inner dump and one external dump. The surface elevation at the southern slope of the mining site ranges from approximately 700 to 703m, with a pit bottom elevation of 615m. Currently, four benches have been formed, each with a height of 10m, and the overall slope angle is 35°. The northern slope of the mining site is adjacent to the Beishan Coal Mine. Currently, four benches have been formed, each with a height of approximately 10 m. The external waste dump is located on the west side of the mining site. Currently, seven dumping benches have been formed, each with a height of 10m, resulting in a total dump height of approximately 90m, with an overall slope angle of 20°. The inner dump is located to the east of the external dump and to the west of the mining area. It currently consists of six dumping benches, each with a height of 20m. The top of the dump site is at an elevation of 760 m, with a total dumping height of approximately 145 m, and the overall maximum slope angle is 22°.

To ensure production safety, the mine is currently equipped with 69 monitoring devices across seven types, including GNSS, crack gauges, and blast vibration sensors. Additionally, five slope radars are deployed, complemented by regularly generated InSAR and UAV topographic data. These instruments have been operating continuously for two years, covering both mining and dumping areas. With multi-source data collected at

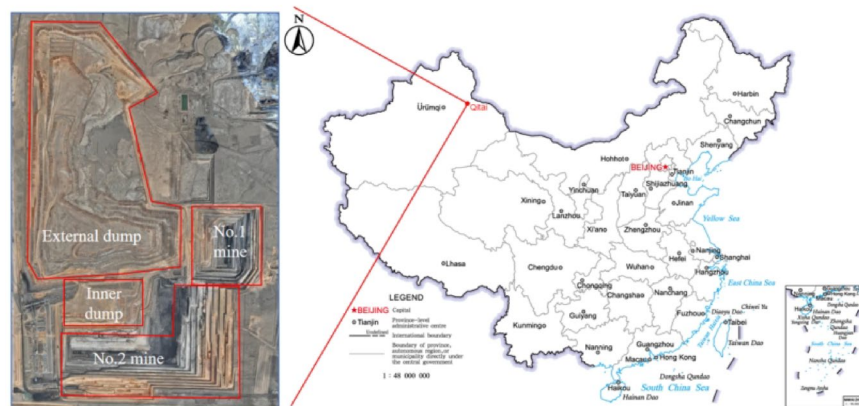


Fig. 6. Study area.

intervals ranging from 5 to 30 min, the system has accumulated approximately 45 million valid data records. Such extensive and long-term data acquisition provides a solid foundation for training and validating the LSTM prediction model and the Dempster–Shafer (D–S) fusion-based risk assessment module.

Evaluation of application performance

(1) VR scenario of an open-pit coal mine

The system enables the integrated visualization of multi-temporal mine models through the development of a UAV model upload interface (Fig. 7a,b). This facilitates intuitive assessment of the current operational status of the open-pit coal mine and detailed analysis of landslide-prone areas. Furthermore, by integrating geological exploration data, a lithological model of the mining stope was developed and visualized using GIS technology (Fig. 7c). This visualization aids users in understanding the ore body's occurrence conditions, thereby supporting the designation of production plans. Additionally, to assist users in optimizing sensor placement based on slope risk assessment, the system incorporates a coordinate point selection function (Fig. 7d). Clicking on the interface allows users to obtain precise location data, providing critical support for determining optimal sensor installation positions.

(2) Multi-source data query

The interface of this module is shown in Fig. 8. This interface enables users to view the quantity and operational status of various devices. The service also provides device querying and management features, allowing users to efficiently view and modify device information as needed.

Simultaneously, the service automatically categorizes received monitoring data by device type, presents detailed information, and provides data querying and adjustment capabilities (Fig. 9).

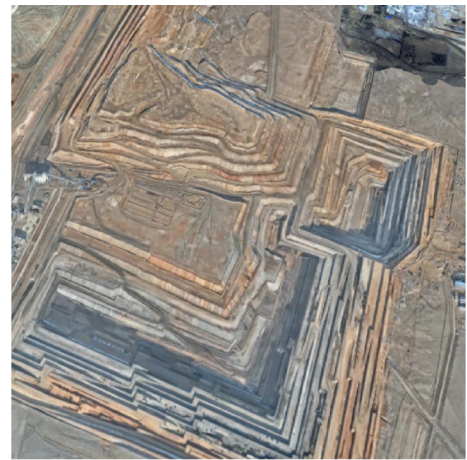
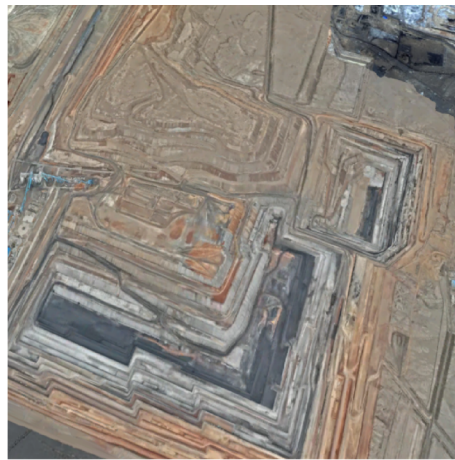
(3) Slope risk prediction.

This functionality is divided into two components: predictive analysis and real-time assessment. The predictive analysis utilizes the aforementioned LSTM model. Taking the received crack meter data as an example, a comparison between predicted and monitored values is shown in Fig. 10. In the figure, the blue line represents the actual monitoring data from January 14 to February 7. The orange line depicts the historical forecast data for the period January 29 to February 7. The green line shows the future forecast data from February 8 to February 17. To evaluate the model's prediction accuracy, the Root Mean Square Error (RMSE) was calculated using the historical predictions (orange line) and the corresponding actual monitoring data for the same period. The calculation results are presented in Table 4. An RMSE of 0.11 indicates high prediction accuracy, demonstrating the reliability of the LSTM algorithm in forecasting mine monitoring data and providing a crucial basis for proactive landslide risk prediction (Table 5).

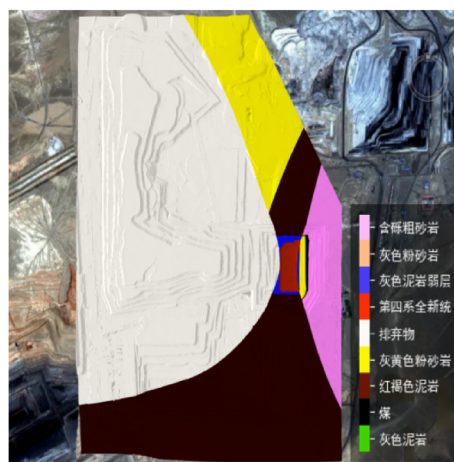
Building on the D–S theory framework described above, both real-time monitoring data and prediction results serve as inputs to identify potential risks of landslides. Using data from Slope Radar #2 on June 25, 2025, as an example, the final calculated risk assessment results are shown in Fig. 11. The results indicate that the vast majority of the monitored area exhibits risk values near 0.01, signifying no immediate risk and a stable condition in these regions.

(4) Multi-Source Data Visualization.

This module provides comprehensive data visualization using a combination of two-dimensional lists, charts, and three-dimensional visualization (Fig. 12). It displays various monitoring data, calculated early-warning



a Open-pit coal model of June 14, 2014 b Open-pit coal model of November 14, 2014



c Lithologic model

d Equipment positioning

Fig. 7. VR scenario of an open-pit coal mine.

Device type	Please select device type		Device name	Please input device name		Q Search	Q reset		
+ New	2 Online	0 Delete	2 Export						
	id	Device type	Device name	Online quantity	Offline quantity	warning quantity	Device pictures	operate	
	43	Geological disaster monitoring	GNSS	49	0	0		2 review	Delete
	45	Geological disaster monitoring	Crack gauge	3	0	0		2 review	Delete
	46	Geological disaster monitoring	Blast vibrometer	3	0	0		2 review	Delete
	47	Geological disaster monitoring	Bidirectional inclinometer	6	6	0		2 review	Delete
	48	Geological disaster monitoring	Pore water pressure	2	2	0		2 review	Delete
	49	Geological disaster monitoring	Multi-point displacement meter	2	2	0		2 review	Delete
	50	Geological disaster monitoring	Bulge instrument	2	0	0		2 review	Delete

Fig. 8. Equipment management services.



Fig. 9. Data management services.

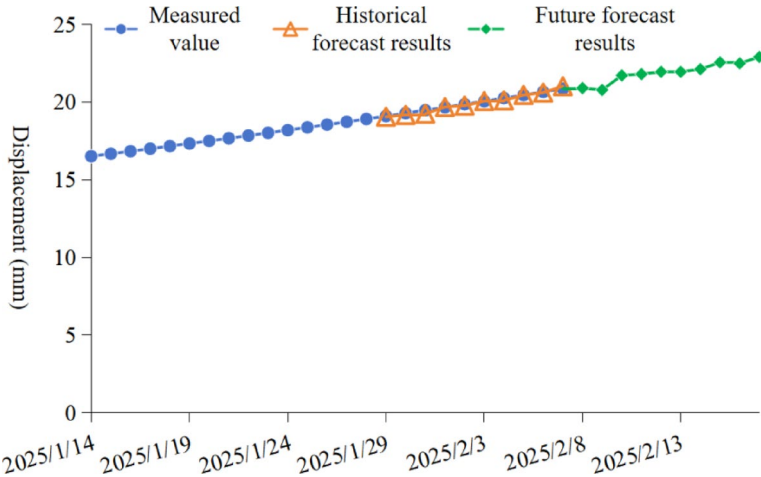


Fig. 10. Displacement prediction results.

Date	Measured data	Predicted data
2025/1/29	19.103	19.055
2025/1/30	19.291	19.180
2025/1/31	19.480	19.249
2025/2/1	19.671	19.690
2025/2/2	19.864	19.776
2025/2/3	20.060	20.064
2025/2/4	20.258	20.109
2025/2/5	20.458	20.458
2025/2/6	20.660	20.614
2025/2/7	20.864	21.023
RMSE	0.1119	

Table 5. Prediction accuracy analysis.

indices, and a live surveillance screen. The interface also enables users to interactively view radar data cloud maps or query historical radar data for selected points, delivering intuitive and comprehensive data insights.

(5) Multi-channel Alert Distribution.

This module generates real-time alerts based on real-time monitoring data and risk assessment results. The module allows configurable response protocols for different warning levels (Fig. 13a), enabling timely intervention by management personnel. Figure 13b demonstrates a blue-level alert response, where a red pop-up window appears in the system interface. Additionally, collected alerts are displayed in a tabular format (Fig. 13c), allowing users to view details and manage responses.

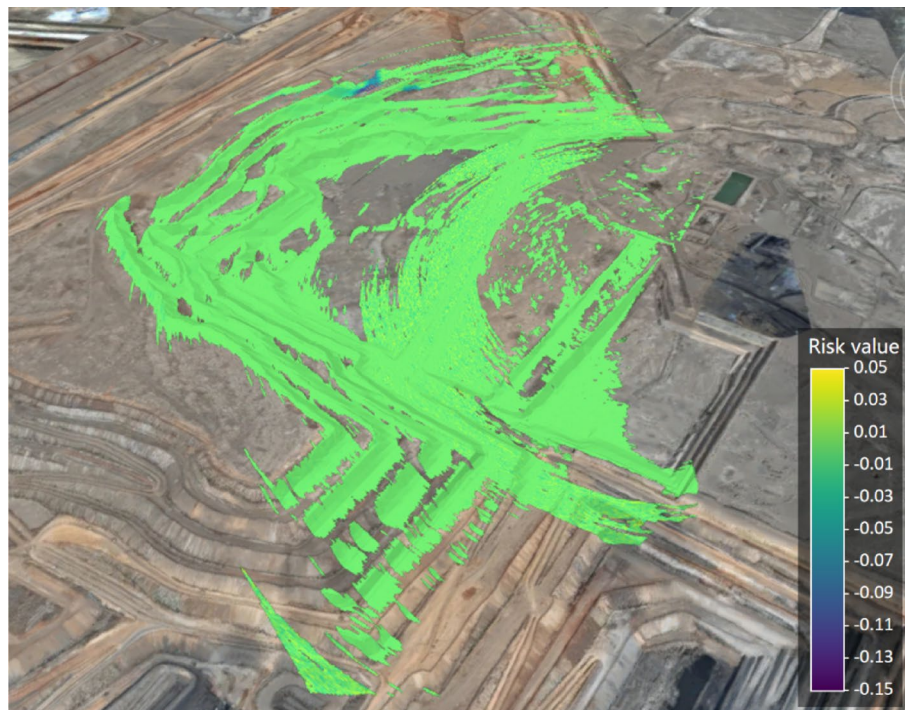


Fig. 11. Real time evaluation results.

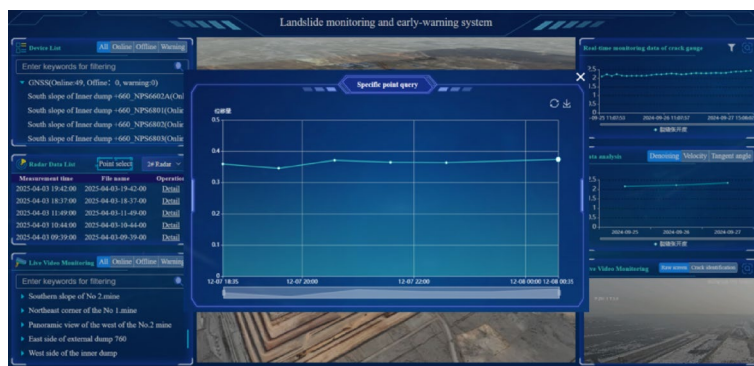


Fig. 12. Data visualization service.

Discussion

Generalizability under different climatic and geological settings

The monitoring and early warning system for slope hazards is designed for long-term, large-scale deployment, typically covering multiple pits and dumps. Taking the Zhonglian Runshi open-pit mine as an example, the monitoring network spans two waste dumps and two mining areas (Fig. 6). These sites exhibit significant geological diversity—ranging from sandstone-dominated benches to clay-rich dumps (Fig. 7c)—and undergo distinct seasonal variations due to climatic influences. Even within the same pit, lithological and structural heterogeneities lead to diverse deformation responses.

The proposed monitoring and early warning method was validated across multiple pits and dumps in the Zhonglian Runshi open-pit mine, which exhibits diverse lithologies and seasonal deformation characteristics. Representative datasets from four zones were used to evaluate the LSTM model's generalization under varying geological and climatic conditions (Table 6). Results show that the model maintained high accuracy ($R^2 > 0.88$) across all seasons and regions.

Prediction accuracy showed slight seasonal variation, being lowest in summer (rainfall period) and recovering in winter. The temporary decline corresponds to rainfall-induced acceleration and increased signal volatility, whereas dry and frozen seasons dominated by stable creep yielded smoother predictions (Fig. 14). Lithology also influenced performance: sandstone benches achieved higher consistency than waste-rock dumps due to

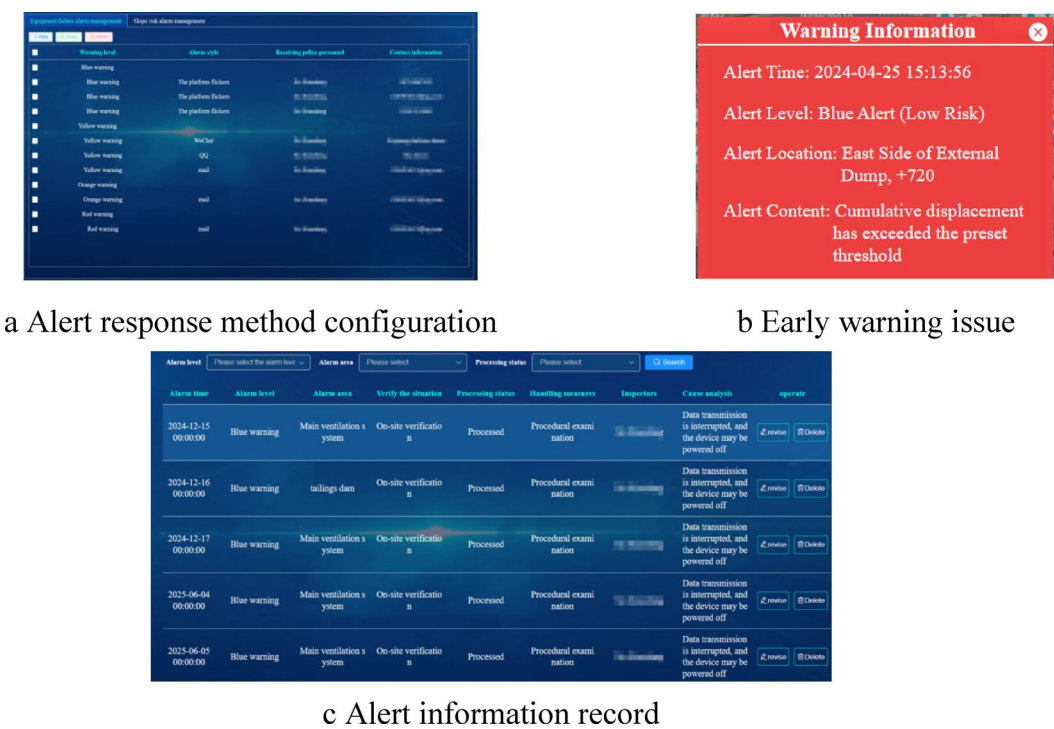


Fig. 13. Schematic diagram of police information release.

Monitoring Zone	Lithology	Season (Time)	Climatic Characteristics	RMSE (mm)	R ²
No.1 Mine—East Bench	Sandstone	Spring (Mar–May)	Snowmelt + Rare Rainfall	0.162	0.93
		Summer (Jun–Aug)	Rainfall	0.214	0.90
		Autumn (Sep–Nov)	Stable, Dry	0.178	0.92
		Winter (Dec–Feb)	Deep Frozen	0.155	0.94
No.2 Mine—South Bench	Mudstone	Spring (Mar–May)	Snowmelt + Rare Rainfall	0.205	0.91
		Summer (Jun–Aug)	Rainfall	0.268	0.88
		Autumn (Sep–Nov)	Stable, Dry	0.221	0.90
		Winter (Dec–Feb)	Deep Frozen	0.189	0.92
Inner Dump—East Slope	Waste rock	Spring (Mar–May)	Snowmelt + Rare Rainfall	0.195	0.91
		Summer (Jun–Aug)	Rainfall	0.232	0.89
		Autumn (Sep–Nov)	Stable, Dry	0.208	0.90
		Winter (Dec–Feb)	Deep Frozen	0.180	0.92
External Dump—South Slope	Waste rock	Spring (Mar–May)	Snowmelt + Rare Rainfall	0.218	0.90
		Summer (Jun–Aug)	Rainfall	0.265	0.88
		Autumn (Sep–Nov)	Stable, Dry	0.230	0.89
		Winter (Dec–Feb)	Deep Frozen	0.201	0.91

Table 6. LSTM prediction performance across regions and seasons.

lower deformation noise. Overall, the LSTM effectively adapted to both long-term creep and short-term rainfall responses, demonstrating strong spatial–temporal generalization for slope deformation forecasting.

Computational and maintenance challenges in long-term deployment

A major advantage of the proposed framework lies in its computational efficiency and operational stability, which make it particularly suitable for real-time deformation monitoring over extended mining areas.

Unlike more complex deep-learning architectures (e.g., BiLSTM or Transformer-based fusion models) that often require dedicated GPUs and frequent retraining, the lightweight LSTM design ensures reliable on-site operation with minimal computational load and maintenance demand.

During a full-year deployment at the Zhonglian open-pit mine. A total of 49 GNSS stations were continuously recorded at a 10-min sampling interval for over 12 months, covering two waste dumps and two mining benches.

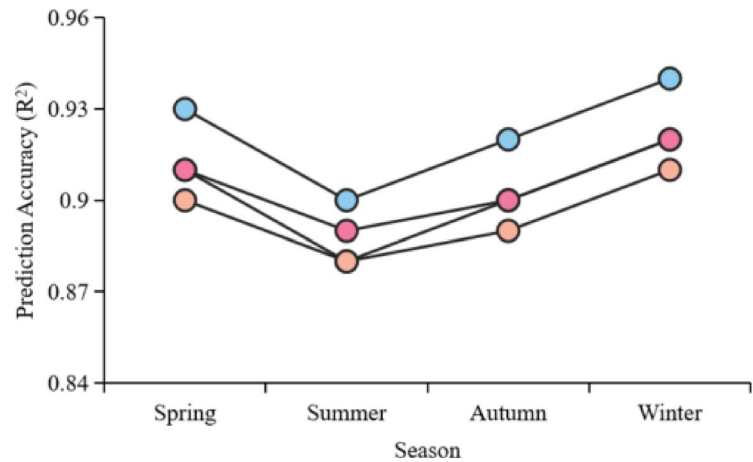


Fig. 14. Prediction accuracy in different seasons.

Period (Month)	Mean Inference Time (s)	CPU Utilization (%)	Memory Usage (GB)	System Downtime (%)
1–3	3.8	24.5	1.2	0.5
4–6	4.2	26.3	1.3	0.6
7–12	4.6	28.1	1.4	0.8

Table 7. Computational performance during 12-month deployment.

Example model	Explanatory Power (R²)	Inference Time (s)	Hardware Demand	Real-time Suitability	Key Limitation
ARMA	0.62	< 1	Very Low	✓✓✓	Fails to capture complex non-linearities
SVM (RBF Kernel)	0.82	1.2	Low	✓✓✓	Performance plateaus with complex temporal sequences
Proposed LSTM	0.91	3.4	Medium	✓✓✓	Optimal balance of accuracy and efficiency for the given hardware
BiLSTM	0.93	5.8	Medium–High	✓✓	Higher latency for a marginal increase in explanatory power
Transformer	0.96	12.5	High	✓	Computationally intensive; requires heavy optimization and data

Table 8. Comparison of real-time prediction performance of algorithms.

The LSTM prediction model runs directly on an industrial PC installed in the field control cabinet, performing fully automated processing and visualization without dependence on remote servers. To evaluate computational stability, all operational logs from the past year were analyzed. The results are summarized in Table 7. Throughout the one-year deployment, the average inference time remained below 0.5 s per update cycle, demonstrating compatibility with the 10-min GNSS sampling rate. CPU utilization and memory consumption remained consistently low and stable, with no evidence of performance degradation or accuracy drift. Slight seasonal increases in computational load were mainly associated with rainfall-induced deformation events that triggered short-term data bursts. Even so, the system sustained smooth operation, confirming that the LSTM structure is sufficiently lightweight for long-term field deployment. The minor downtime (<1%) was primarily due to scheduled firmware maintenance rather than algorithmic or hardware instability.

Comparison with alternative methods

The selection of an algorithm for a real-time deformation prediction necessitates a multi-faceted evaluation, where predictive fidelity must be carefully weighed against computational efficiency and operational constraints. To inform this decision, we conducted a systematic benchmark of representative models spanning diverse paradigms, including traditional statistical methods (ARMA), classical machine learning (SVM), and contemporary deep learning architectures (BiLSTM, Transformer). The quantitative outcomes as shown in Table 8.

The results highlight a critical trade-off in model selection for real-time early warning: higher accuracy often comes at the cost of speed and computational demand. Simple models like ARMA are fast but lack the accuracy ($R^2=0.62$) required for reliable predictions. On the other hand, highly complex models like the Transformer achieve top-tier accuracy ($R^2=0.96$), but their high computational cost leads to slow inference times (12.5s).

Our proposed LSTM model offers the best compromise. It delivers high accuracy ($R^2=0.91$), closely approaching the performance of more complex models, while maintaining a reasonable inference time (3.4s) suitable for real-time processing. This balance is crucial for the Zhonglian Runshi mine, where server capacity is limited. Therefore, the LSTM is the optimal choice, providing a robust and feasible solution for 24/7 automated monitoring without requiring expensive hardware upgrades.

Advantages and disadvantages of the EWS

EWS exhibit a strong dependency on the quality of monitoring data. In the challenging environments typical of mine sites, a diverse array of monitoring instruments is deployed. However, the data from these varied sources inevitably present challenges, including noise contamination, data gaps, and complex interrelationships. During the data preprocessing stage in this study, the application of the Pauta criterion effectively removed a significant portion of noisy data, laying a foundation for subsequent analysis. Nevertheless, during extreme weather events or sudden geological hazards, monitoring data may contain abrupt anomaly signals reflecting the rapid destabilization of the slope. The Pauta criterion, relying on fixed statistical thresholds derived from historical data, struggles to differentiate these critical anomalies from common noise, posing a risk of inadvertently removing crucial warning signals. Consequently, future research should focus on developing more intelligent, adaptive data preprocessing algorithms. These algorithms should dynamically adjust filtering criteria based on real-time data characteristics, aiming to maximize the retention of potential critical risk information while effectively removing noise, thereby enhancing the overall reliability and effectiveness of the monitoring and warning system under extreme conditions.

Furthermore, while the stability of mine slopes is influenced by multifaceted factors, most existing monitoring and warning methods predominantly focus on a single data type. The model developed in this study primarily analyzes displacement-related data, which, while capturing certain deformation characteristics, underutilizes data from other commonly deployed instruments such as rain gauges and piezometers. This limitation introduces a degree of partiality into the risk assessment. Future work must therefore explore more comprehensive data fusion algorithms. Such algorithms should facilitate information synergy between different monitoring data types, enabling the construction of an integrated risk warning model that incorporates multi-dimensional data—including displacement, stress, and environmental factors. This holistic approach is essential for achieving more precise and comprehensive dynamic monitoring and early warning of slope risks, ultimately providing more robust decision-making support for the safe operation of open-pit coal mines.

Conclusion

This study addresses critical challenges in open-pit coal mine slope monitoring and early warning systems, including data silos and "prioritizing data acquisition over analysis." By integrating IoT, cloud computing, deep learning, and GIS technologies, we developed a cloud-based system for slope risk monitoring and early warning in open-pit coal mines. The system underwent preliminary implementation at the Zhonglian Runshi Open-Pit Coal Mine, yielding the following conclusions:

- (1) The system leverages multi-source monitoring devices to achieve efficient measurement and real-time transmission of slope surface displacement, internal stress, and environmental parameters. Through an LSTM time-series prediction model and multi-indicator risk fusion evaluation based on D-S evidence theory, it significantly enhances dynamic risk perception and advanced early-warning capabilities for slope stability.
- (2) Designed with a B/S architecture, the system integrates modules including: VR scenario of open-pit coal mine, multi-source data query, slope risk prediction, multi-source data visualization, and Multi-channel alert distribution. This establishes an end-to-end closed-loop management framework spanning data acquisition, analysis, and decision support. Field validation at the Zhonglian Runshi Mine demonstrated the system's high accuracy and reliability in slope deformation prediction and risk classification. These EWS offer a valuable tool for effective landslide monitoring, early warning, and risk prevention in open-pit mining areas.

Data availability

The data that support the findings of this study are available on request from the corresponding author, Yang Qiu, upon reasonable request.

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Author contributions

Methodology, Xiaoyong Dong; software, Shuang Li and Yang Qiu; data curation, Ran Ma; writing original draft preparation, Weihui Tian. Yang Qiu; writing-review and editing, Hui Xiang; supervision, Jiongjiong Zhu; funding acquisition, Keke Zhao. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no conflict of interest.

Additional information

Correspondence and requests for materials should be addressed to R.M. or Y.Q.

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